Machine Learning
Lecture 8

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Probabilities (recap)

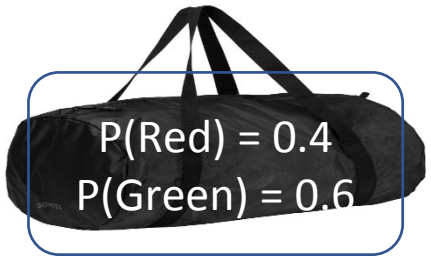
Probabilities (Recap)

- $P(X = 0) =$
- $P(Y = 3) =$
- $P(X = 1, Y = 2) =$
- $P(Y = 2, X = 1) =$
- $P(X = 1 \mid Y = 2) =$
- $P(Y = 2 \mid X = 1) =$

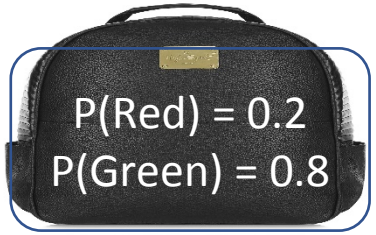
X	Y
0	0
0	1
1	0
1	2
2	3
2	0
2	3
1	3
1	2
0	3
0	2
0	0

Probabilities (Recap)

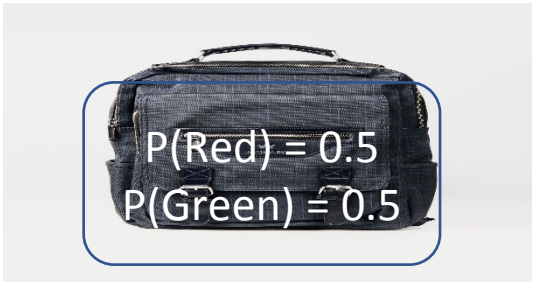
P(bag1) = 0.3



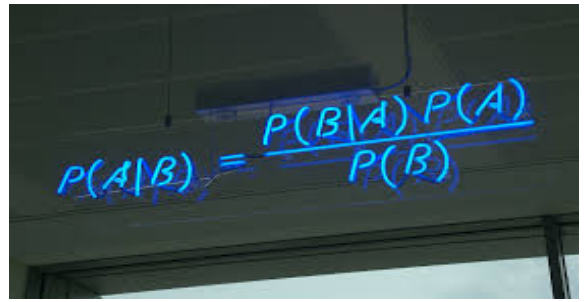
P(bag2) = 0.5



P(bag3)=0.2



- P(bag2) =
- P(Red|bag1) =
- P (Red) =
- P(bag1|Red) =



Probabilities (Recap)

- $P(A, B)$ or $P(A \cap B) = ?$
- $P(A, B) = P(A \mid B) * P(B) = P(B \mid A) * P(A)$

- $P(A, B \mid C) = P(A \mid B, C) * P(B \mid C) = P(B \mid A, C) * P(A)$

- $P(A \cap B \cap C) = P(A, B, C)$

- $P(A \cap B \cap C \cap D) = ?$

Probabilities (Recap)

if all events are independent

- P (A , B) = P(A) * P (B)
- P (A , B , C) =
- P (A , B , C, D) =
- P(A1,A2, . . ., An) =

Conditionally independent

- P (A , B | C) = P(A | C) * P (B | C)
- P (A , B , C | D) =
- P(A1,A2, . . ., An | Z) =



Q & A



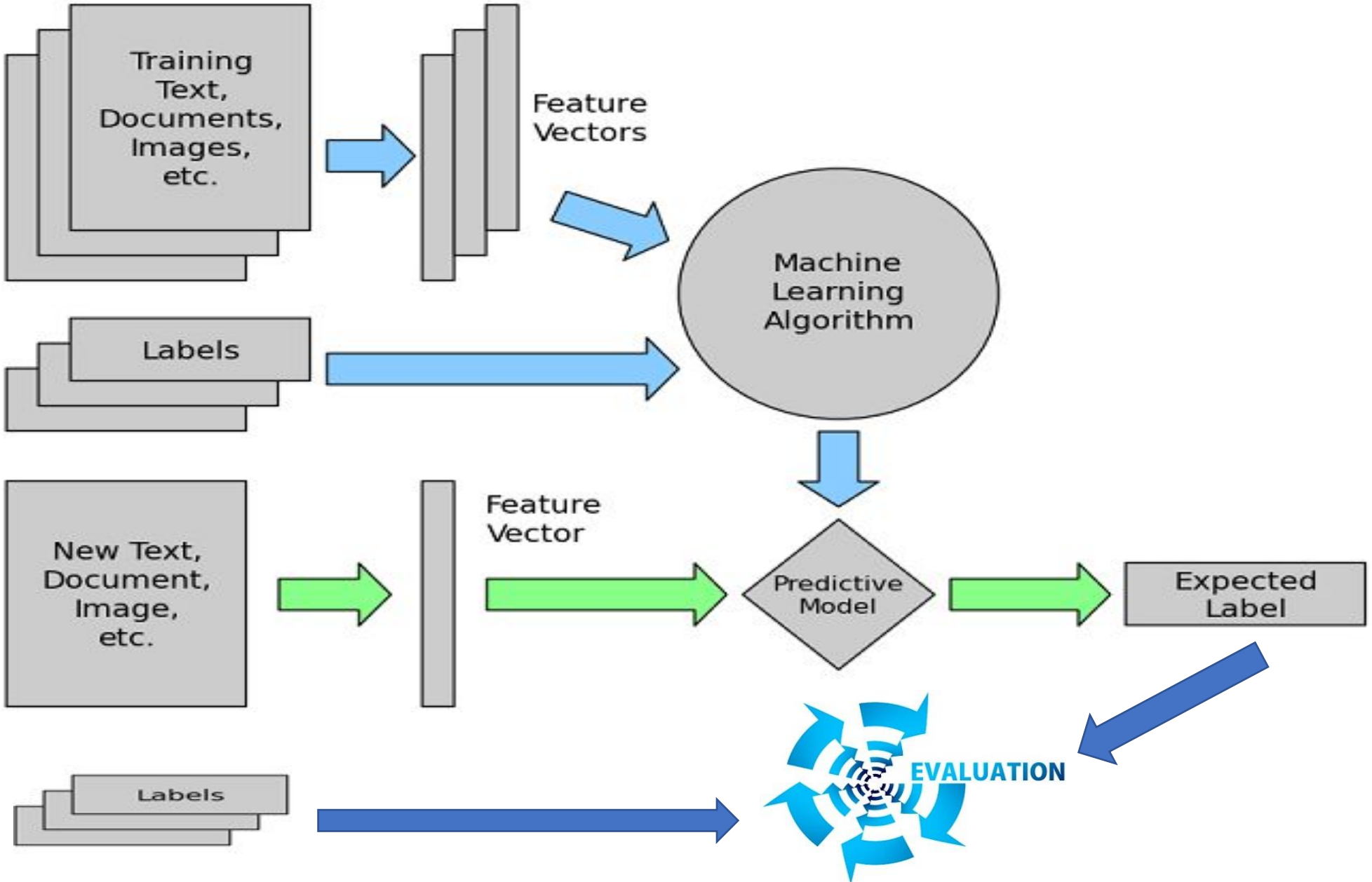
Naïve Bayes Classifier

(Classification Algorithms)

Naïve Bayes Classifier

Training Data

Testing Data



Naïve Bayes Classifier

- Assumption

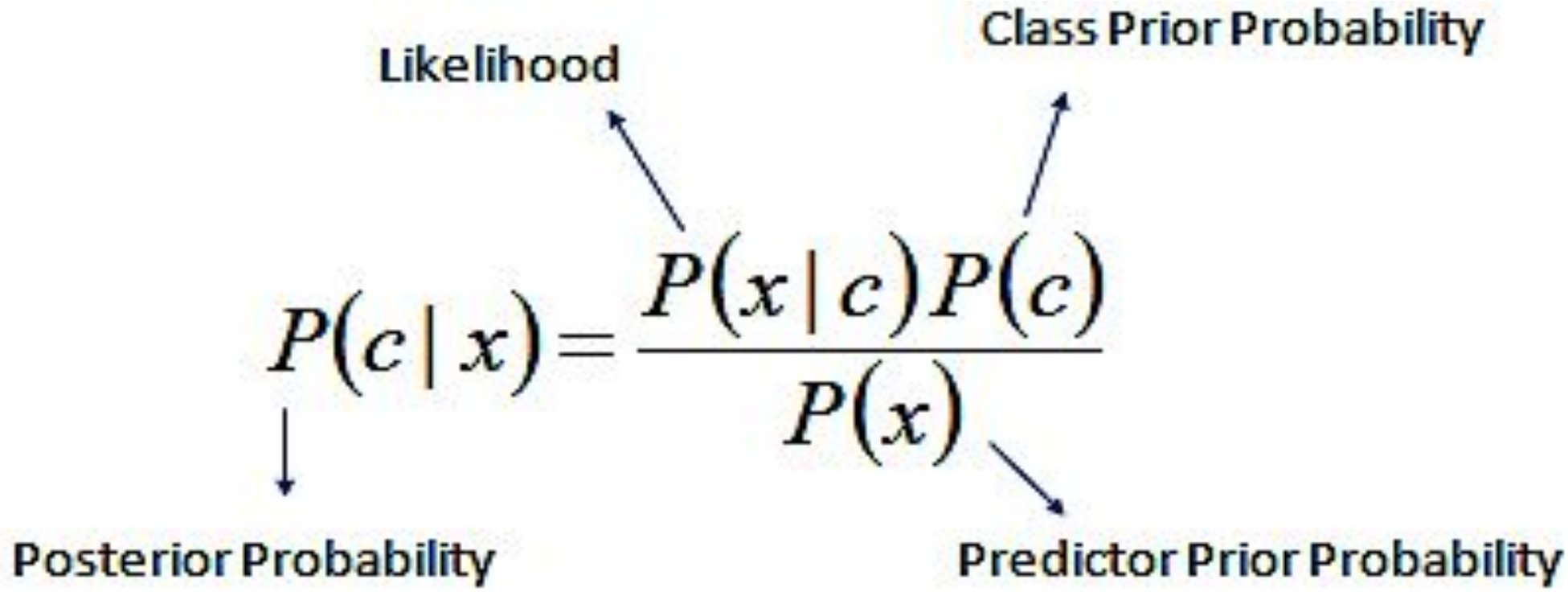
Conditional Independence: Assume the feature probabilities x are independent given the class *c.*

$$P(x \mid c)$$

$$P(x_1, x_2, \ldots, x_n \mid c)$$

$$P(x_1, \ldots, x_n \mid c) = P(x_1 \mid c) \bullet P(x_2 \mid c) \bullet P(x_3 \mid c) \bullet \ldots \bullet P(x_n \mid c)$$

Naïve Bayes Classifier



$$P(c \mid X) = P(x_1 \mid c) \times P(x_2 \mid c) \times \cdots \times P(x_n \mid c) \times P(c)$$

Naïve Bayes Classifier

Bayes Rule: $$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

$$c_{MAP} = \underset{c \in C}{\operatorname{argmax}}\, P(c \mid x)$$

MAP is "maximum a posteriori" = most likely class

$$= \underset{c \in C}{\operatorname{argmax}}\, \frac{P(x \mid c)P(c)}{P(x)}$$

Bayes Rule

$$= \underset{c \in C}{\operatorname{argmax}}\, P(x \mid c)P(c)$$

Dropping the denominator

Naïve Bayes Classifier

- Training Data

x1	x2	x3	x4	x5	x6	Class
1	0	0	1	1	1	y
1	1	0	1	1	1	y
0	1	0	0	0	0	y
0	0	1	0	0	1	n
1	1	0	1	1	0	n
0	1	0	1	1	0	n

P(y) = P(n) =

P(x1=1|y) = P(x1=0|y)=

P(x1=1|n) = P(x1=0|n) =

P(x2=1|y) = P(x2=0|y) =

P(x2=1|n) = P(x2=0|n) =

P(x3=1|y) = P(x3=0|y) =

P(x3=1|n) = P(x3=0|n) =

P(x4=1|y) = P(x4=0|y) =

P(x4=1|n) = P(x4=0|n) =

P(x5=1|y) = P(x5=0|y) =

P(x5=1|n) = P(x5=0|n) =

P(x6=1|y) = P(x6=0|y) =

P(x6=1|n) = P(x6=0|n) =

- Testing Data

1, 1, 0, 0, 0, 1 y

1, 0, 1, 1, 1, 1 y

$$\underset{c \in C}{\operatorname{argmax}} P(x \mid c) P(c)$$

Naïve Bayes Classifier

- Training Data

x1	x2	x3	x4	x5	x6	Class
1	0	0	1	1	1	y
1	1	0	1	1	1	y
0	1	0	0	0	0	y
0	0	1	0	0	1	n
1	1	0	1	1	0	n
0	1	0	1	1	0	n

P(y) = 1/2 P(n) = 1/2

P(x1=1|y) =2/3 P(x1=0|y) =1/3

P(x1=1|n) = 1/3 P(x1=0|n) = 2/3

P(x2=1|y) = 2/3 P(x2=0|y) = 1/3

P(x2=1|n) = 2/3 P(x2=0|n) = 1/3

P(x3=1|y) = 0 P(x3=0|y) = 1

P(x3=1|n) = 1/3 P(x3=0|n) = 2/3

P(x4=1|y) = 2/3 P(x4=0|y) = 1/3

P(x4=1|n) = 2/3 P(x4=0|n) = 1/3

P(x5=1|y) = 2/3 P(x5=0|y) = 1/3

P(x5=1|n) = 2/3 P(x5=0|n) = 1/3

P(x6=1|y) = 2/3 P(x6=0|y) = 1/3

P(x6=1|n) = 1/3 P(x6=0|n) = 2/3

- Testing Data

1, 1, 0, 0, 0, 1 y

1, 0, 1, 1, 1, 1 y

$$\underset{c \in C}{\operatorname{argmax}} P(x \mid c) P(c)$$

Naïve Bayes Classifier

- Training Data

x1	x2	x3	x4	x5	x6	Class
1	0	0	1	1	1	y
1	1	0	1	1	1	y
0	1	0	0	0	0	y
0	0	1	0	0	1	n
1	1	0	1	1	0	n
0	1	0	1	1	0	n

P(y) = 1/2 P(n) = 1/2

P(x1=1|y) =2/3 P(x1=0|y) =1/3

P(x1=1|n) = 1/3 P(x1=0|n) = 2/3

P(x2=1|y) = 2/3 P(x2=0|y) = 1/3

P(x2=1|n) = 2/3 P(x2=0|n) = 1/3

P(x3=1|y) = 0 P(x3=0|y) = 1

P(x3=1|n) = 1/3 P(x3=0|n) = 2/3

P(x4=1|y) = 2/3 P(x4=0|y) = 1/3

P(x4=1|n) = 2/3 P(x4=0|n) = 1/3

P(x5=1|y) = 2/3 P(x5=0|y) = 1/3

P(x5=1|n) = 2/3 P(x5=0|n) = 1/3

P(x6=1|y) = 2/3 P(x6=0|y) = 1/3

P(x6=1|n) = 1/3 P(x6=0|n) = 2/3

- Testing Data

1, 1, 0, 0, 0, 1 y

1, 0, 1, 1, 1, 1 y

$$\underset{c \in C}{\operatorname{argmax}} P(x \mid c) P(c)$$



Q & A

Naïve Bayes Classifier (smoothing)

Naïve Bayes Classifier (Smoothing)

- A solution would be **Laplace smoothing** , which is a technique for smoothing categorical data.
- A small-sample correction, or **pseudo-count**, will be incorporated in every probability estimate.
- Consequently, no probability will be zero.
- This is a way of regularizing Naive Bayes, and when the pseudo-count is zero, it is called Laplace smoothing.
- While in the general case it is often called **Lidstone smoothing**.

Naïve Bayes Classifier (after Smoothing)

- Training Data

x1	x2	x3	x4	x5	x6	Class
1	0	0	1	1	1	y
1	1	0	1	1	1	y
0	1	0	0	0	0	y
0	0	1	0	0	1	n
1	1	0	1	1	0	n
0	1	0	1	1	0	n

P(y) = P(n) =
P(x1=1|y) = P(x1=0|y)=
P(x1=1|n) = P(x1=0|n) =
P(x2=1|y) = P(x2=0|y) =
P(x2=1|n) = P(x2=0|n) =
P(x3=1|y) = P(x3=0|y) =
P(x3=1|n) = P(x3=0|n) =
P(x4=1|y) = P(x4=0|y) =
P(x4=1|n) = P(x4=0|n) =
P(x5=1|y) = P(x5=0|y) =
P(x5=1|n) = P(x5=0|n) =
P(x6=1|y) = P(x6=0|y) =
P(x6=1|n) = P(x6=0|n) =

- Testing Data

1, 1, 0, 0, 0, 1 y
1, 0, 1, 1, 1, 1 y

$$\underset{c \in C}{\operatorname{argmax}} P(x \mid c) P(c)$$

Naïve Bayes Classifier (Smoothing)

- Training Data

x1	x2	x3	x4	x5	x6	Class
1	0	0	1	1	1	y
1	1	0	1	1	1	y
0	1	0	0	0	0	y
0	0	1	0	0	1	n
1	1	0	1	1	0	n
0	1	0	1	1	0	n

P(y) = 1/2 P(n) = 1/2

P(x1=1|y) =3/5 P(x1=0|y) =2/3

P(x1=1|n) = 2/5 P(x1=0|n) = 3/5

P(x2=1|y) = 3/5 P(x2=0|y) = 2/5

P(x2=1|n) = 3/5 P(x2=0|n) = 2/5

P(x3=1|y) = 1/5 P(x3=0|y) = 4/5

P(x3=1|n) = 2/5 P(x3=0|n) = 3/5

P(x4=1|y) = 3/5 P(x4=0|y) = 2/5

P(x4=1|n) = 3/5 P(x4=0|n) = 2/5

P(x5=1|y) = 3/5 P(x5=0|y) = 2/5

P(x5=1|n) = 3/5 P(x5=0|n) = 2/5

P(x6=1|y) = 3/5 P(x6=0|y) = 2/5

P(x6=1|n) = 2/5 P(x6=0|n) = 3/5

- Testing Data

1, 1, 0, 0, 0, 1 y

1, 0, 1, 1, 1, 1 y

$$\underset{c \in C}{\operatorname{argmax}} P(x \mid c) P(c)$$



Q & A



Programming Assignment (example)

Naïve Bayes Classifier

- Training Data

x1	x2	x3	x4	x5	x6	Class
1	0	0	1	1	1	y
1	1	0	1	1	1	y
0	1	0	0	0	0	y
0	0	1	0	0	1	n
1	1	0	1	1	0	n
0	1	0	1	1	0	n

P(y) = P(n) =

P(x1=1|y) = P(x1=0|y)=

P(x1=1|n) = P(x1=0|n) =

P(x2=1|y) = P(x2=0|y) =

P(x2=1|n) = P(x2=0|n) =

P(x3=1|y) = P(x3=0|y) =

P(x3=1|n) = P(x3=0|n) =

P(x4=1|y) = P(x4=0|y) =

P(x4=1|n) = P(x4=0|n) =

P(x5=1|y) = P(x5=0|y) =

P(x5=1|n) = P(x5=0|n) =

P(x6=1|y) = P(x6=0|y) =

P(x6=1|n) = P(x6=0|n) =

- Testing Data

1, 1, 0, 0, 0, 1 y

1, 0, 1, 1, 1, 1 y

$$\underset{c \in C}{\operatorname{argmax}} P(x \mid c) P(c)$$

Naïve Bayes Classifier

x1	x2	x3	x4	x5	x6	Class
1	0	0	1	1	1	y
1	1	0	1	1	1	y
0	1	0	0	0	0	y
0	0	1	0	0	1	n
1	1	0	1	1	0	n
0	1	0	1	1	0	n

```
X_Train = np.array([[1,0,0,1,1,1],
                    [1,1,0,1,1,1],
                    [0,1,0,0,0,0],
                    [0,0,1,0,0,1],
                    [1,1,0,1,1,0],
                    [0,1,0,1,1,0]])
```

```
print(X_Train)
```

```
1  yes_train = X_Train[:3,:]
2  print(yes_train)
```

```
[[1 0 0 1 1 1]
 [1 1 0 1 1 1]
 [0 1 0 0 0 0]]
```

```
yes_train.shape
```

```
1  yes_one_Prob = yes_train.sum(axis=0)/yes_train.shape[0]
2  print(yes_one_Prob)
```

```
[0.66666667 0.66666667 0.         0.66666667 0.66666667 0.66666667]
```

```
1  yes_zero_Prob = 1 - yes_one_Prob
2  print(yes_zero_Prob)
```

```
[0.33333333 0.33333333 1.         0.33333333 0.33333333 0.33333333]
```

Naïve Bayes Classifier

- Training Data

x1	x2	x3	x4	x5	x6	Class
1	0	0	1	1	1	y
1	1	0	1	1	1	y
0	1	0	0	0	0	y
0	0	1	0	0	1	n
1	1	0	1	1	0	n
0	1	0	1	1	0	n

P(y) = 1/2 P(n) = 1/2

P(x1=1|y) =2/3 P(x1=0|y) =1/3

P(x1=1|n) = 1/3 P(x1=0|n) = 2/3

P(x2=1|y) = 2/3 P(x2=0|y) = 1/3

P(x2=1|n) = 2/3 P(x2=0|n) = 1/3

P(x3=1|y) = 0 P(x3=0|y) = 1

P(x3=1|n) = 1/3 P(x3=0|n) = 2/3

P(x4=1|y) = 2/3 P(x4=0|y) = 1/3

P(x4=1|n) = 2/3 P(x4=0|n) = 1/3

P(x5=1|y) = 2/3 P(x5=0|y) = 1/3

P(x5=1|n) = 2/3 P(x5=0|n) = 1/3

P(x6=1|y) = 2/3 P(x6=0|y) = 1/3

P(x6=1|n) = 1/3 P(x6=0|n) = 2/3

- Testing Data

1, 1, 0, 0, 0, 1 y

1, 0, 1, 1, 1, 1 y

$$\underset{c \in C}{\operatorname{argmax}} P(x \mid c) P(c)$$



Q & A

Programming Assignment (OCR)

Programming Assignment

- Implement an OCR system that distinguishes between the images of digit 4 and digit 2.
- Four files (trainX, trainY, testX and testY)

```
%matplotlib inline
import numpy as np
from matplotlib import pyplot as plt
```

```
img2=np.array([0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,1,1,0,0,0,0,0,0,0,0,
               0,0,0,0,0,1,1,0,0,0,0,0,0,0,0,0,0,1,0,0,1,0,0,0,0,0,0,0,0,
               0,0,0,0,1,1,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,1,1,0,0,1,0,0,0,
               0,0,0,0,0,0,0,0,1,1,1,0,1,0,0,0,0,0,0,0,0,0,0,1,1,0,1,0,
               1,0,0,0,0,0,0,0,0,0,0,0,1,0,0,1,0,1,0,0,0,0,0,0,0,0,0,0,1,0,
               0,0,1,0,1,1,1,1,1,1,1,1,1,1,1,0,0,0,1,1,0,0,0,0,0,0,0,0,0,
               0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,
               0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,
               0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,
               0,0,0,0],dtype=np.uint8)
```

Programming Assignment

```
1 img2.shape
```

```
(256,)
```

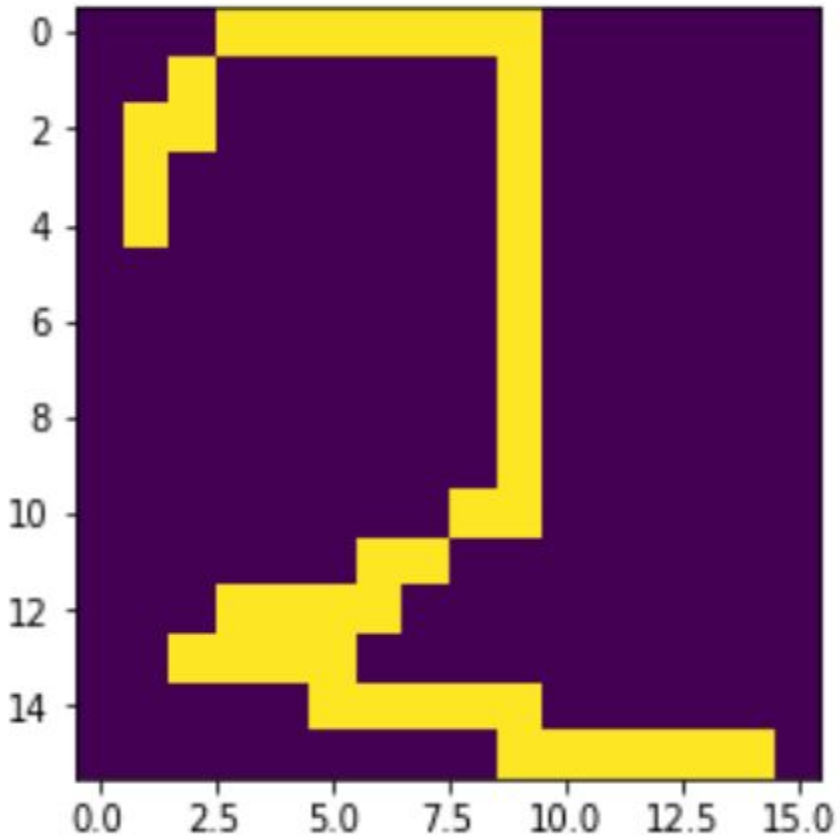
```
img = np.reshape(img2,(16,16),order='F')
```

```
1 img.shape
```

```
(16, 16)
```

```
1 plt.imshow(img)
```

```
<matplotlib.image.AxesImage at 0x5cc1780>
```





Q & A